

WHEELING OHIO COUNTY AP, WV, USA

WMO: 724275

Lat: 40.176N

Lon: 80.647W

Elev: 1179

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	9.8	-3.8	4.7	8.3	0.2	5.8	12.2	25.4	38.3	23.2	35.6	8.0	260	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	88.2	72.4	85.7	71.3	83.6	70.1	75.2	83.9	73.7	81.8	72.5	80.1	7.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	125.5	79.8	71.3	120.8	78.5	70.1	115.6	77.1	39.3	84.3	38.0	82.1	36.8	79.8	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.7	16.7	0.3	90.9	5.3	3.1	-3.5	93.2	-6.6	95.0	-9.6	96.8	-13.4	99.1		
			-0.6	77.8	5.1	1.7	-4.3	79.0	-7.3	79.9	-10.1	80.9	-13.9	82.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	29.3	32.2	40.5	52.8	62.0	68.9	72.8	71.8	65.8	54.5	43.9	34.5		
	DBStd	17.83	12.41	11.44	11.38	9.59	8.02	5.97	4.96	4.86	6.94	9.03	9.79	10.69		
	HDD50	2374	652	509	341	80	7	0	0	0	0	56	235	494		
	HDD65	5360	1108	918	760	380	155	31	4	6	72	345	635	946		
	CDD50	3294	9	11	46	163	380	568	708	676	475	195	50	13		
	CDD65	807	0	0	1	13	63	150	247	217	97	18	1	0		
	CDH74	5806	0	1	9	110	488	1038	1830	1523	701	103	3	0		
	CDH80	1498	0	0	0	12	86	248	542	414	179	17	0	0		
	WSAvg	7.7	9.6	9.3	9.1	9.1	7.1	6.5	5.7	5.4	5.9	7.3	8.4	9.0		
	PrecAvg	39.9	3.2	2.5	3.5	3.5	4.2	4.1	4.1	3.5	3.2	2.6	2.7	2.9		
Precipitation	PrecMax	53.7	7.1	5.4	5.8	4.7	6.2	9.6	8.1	6.4	9.3	5.7	4.2	6.4		
	PrecMin	31.2	1.5	1.3	1.3	1.6	1.3	1.2	2.6	1.6	1.1	0.7	0.6	1.3		
	PrecStd	5.6	1.5	1.0	1.3	0.9	1.5	1.7	1.4	1.5	1.8	1.3	0.9	1.3		
	PrecStd	5.6	1.5	1.0	1.3	0.9	1.5	1.7	1.4	1.5	1.8	1.3	0.9	1.3		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	67.1	74.3	81.7	85.7	88.4	91.2	90.7	88.7	82.4	71.8	64.4		
		MCWB	55.4	55.2	58.4	62.4	69.0	71.7	74.9	72.4	70.9	68.5	59.7	57.0		
	2%	DB	58.4	60.9	69.4	77.3	82.2	85.3	88.2	87.6	84.8	76.5	66.8	59.8		
		MCWB	52.9	51.7	55.4	59.6	67.0	70.8	73.3	72.4	69.8	64.2	56.3	53.7		
	5%	DB	53.4	55.1	64.0	73.1	79.6	82.9	85.4	84.3	81.4	72.6	63.1	55.4		
		MCWB	49.3	48.7	52.5	58.0	66.1	70.2	72.1	70.7	68.7	62.0	54.2	50.0		
	10%	DB	47.2	49.8	58.8	69.2	76.2	80.6	82.8	81.7	78.2	69.1	59.1	51.0		
		MCWB	42.7	42.9	50.1	55.6	64.3	69.1	70.8	69.6	66.7	60.8	52.0	46.0		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.1	57.9	60.6	65.3	72.5	75.2	77.7	76.7	75.0	70.7	62.4	59.1		
		MCDB	62.0	65.3	71.0	76.7	82.6	83.7	87.3	86.1	84.3	79.3	68.5	63.4		
	2%	WB	53.9	53.8	57.5	62.3	69.7	73.3	75.6	74.7	72.5	67.5	58.8	55.1		
		MCDB	57.8	58.6	66.2	73.0	79.0	82.0	84.5	82.9	80.5	73.3	64.7	58.6		
	5%	WB	49.4	49.0	54.9	60.2	67.9	71.9	74.1	73.2	70.6	64.4	55.7	51.2		
		MCDB	52.7	55.8	62.1	69.0	76.4	80.4	82.1	80.4	77.4	69.9	61.1	54.9		
	10%	WB	43.0	43.6	51.0	58.1	66.0	70.5	72.6	71.7	68.8	61.3	52.7	46.1		
		MCDB	47.0	49.4	57.9	66.2	74.0	78.1	80.0	78.5	75.1	67.8	58.4	50.0		
Mean Daily Temperature Range	5% DB	MDBR	13.7	14.9	17.4	19.6	18.9	17.7	17.7	18.1	18.7	17.1	15.3	12.8		
		MCDBR	18.2	21.5	23.4	23.8	22.0	19.9	19.9	20.7	21.6	21.0	20.1	18.2		
		MCWBR	15.0	16.5	14.1	13.0	11.4	9.6	8.2	8.5	9.7	11.3	13.5	14.9		
	5% WB	MCDBR	17.4	21.0	20.3	20.2	18.8	17.1	17.2	16.6	17.4	17.8	17.4	17.9		
		MCWBR	15.6	17.8	14.0	12.9	10.8	9.5	8.3	8.2	8.8	11.0	13.8	16.4		
Clear-Sky Solar Irradiance	taub		0.301	0.322	0.347	0.381	0.423	0.452	0.462	0.460	0.394	0.361	0.318	0.299		
	taud		2.473	2.393	2.397	2.330	2.221	2.173	2.167	2.153	2.317	2.413	2.512	2.518		
	Ebn at Noon		275	284	287	283	272	263	259	256	267	265	266	268		
	Edh at Noon		24	30	34	39	44	47	47	46	36	30	23	22		
All-Sky Solar Radiation	RadAvg		491	762	1074	1448	1688	1836	1821	1671	1367	922	626	422		
	RadStd		49	112	97	139	139	144	123	95	134	94	85	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+112

Nomenclature: See separate page